ISA5810 DM2026 Lab 1 — Master Notebook Questions

Fill in this file as you work through DM2026-Lab1-Master.ipynb. Submit this completed file converted to PDF together with the rest of your Lab 1 work.

Please do not change the style of this document.

Question numbers 1–10 follow the same order as the notebook (top to bottom).

Student Information

Please complete the fields below. For **English Name** and **Chinese Name**, you may fill in **only one** if you do not have the other.

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Instructions

These questions go with the Master Jupyter notebook (DM2026-Lab1-Master.ipynb). Each question is tied to a specific notebook section. Open the notebook, find **Question N (take home)**, and write your answer here.

We **strongly** suggest running the notebook to learn the lab, but the TAs will **not** audit whether you executed every cell to answer the questions in this document.

How to answer each question

1. Find the **notebook section** listed under the question (use the Table of Contents if needed).
2. Read the task; change parameters when asked (e.g. stop_words, thresholds, k, minSup).
3. Write a clear answer in the **Answer** space below each question.

4. Leave the question text unchanged; write only in the answer and disclosure fields.

Sources, honesty, and AI use

Acceptable vs. Unacceptable AI Usage: Generative AI should serve as an interactive tutor to guide your understanding, not as an automated solver completing the assignment for you.

- **Permitted (AI as Guide & Editor):**
  - Asking AI to explain unfamiliar theoretical concepts, discuss algorithmic trade-offs, or troubleshoot runtime errors in the notebook.
  - Polishing the grammar, phrasing, and readability of your own original, self-written drafts.

- **Prohibited (AI as Solver):**
  - Copying assignment questions directly into an AI tool to generate answers.
  - Using AI to fabricate experimental results or observations without executing the code in DM2026-Lab1-Master.ipynb.
  - Submitting machine-generated prose wholesale.

- **External material:** If you use websites, books, papers, or documentation beyond the notebook, cite them with **clickable links** (or DOI/ISBN when no URL exists).

- **When AI was used:** Your answer is acceptable if you (1) **disclose** AI use and (2) include **at least one external link** that validates the information in your answer (official docs, textbook, paper, or reputable tutorial — not the AI chat alone as the only source).

- **When AI was not used:** You may write "No" under **AI used** or leave it blank only if you truly did not use AI for that question. References are still encouraged if you looked anything up.

Under each question, complete:

- **AI used for this answer:** Yes / No
- **If yes, how:** (e.g. brainstorming, grammar, concept explanation)

- **External references (links):** *(required when AI = Yes; encouraged otherwise)*

Write in English. Keep answers concise and specific.

Questions

Question 1: Data Filtering & Domain Relevance

Notebook: Section 3.2 — Question 1 (take home)

Task: Filter the dataframe **X** for the category **sci.med** and check its length.

Questions:

1. How many documents are in this specific category?
2. From a data mining perspective, why is isolating specific categories a crucial first step before applying algorithms like frequent pattern mining?

Answer:

1. *There are 594 documents*

2. *Because, every category has different kind of features, so if we mixed them, it would causes three problems. First, support is diluted: a word set that is common inside sci.med may fall below minSup when it is counted over all 2,257 documents. Second, the top patterns become generic words shared by every topic, so the result is biased toward that category. Third, a pattern mined inside one category can be linked to that topic, so it can be interpreted and used as a feature.*

AI used for this answer: No

Question 2: Handling Missing Data

Notebook: Section 4.1 — Question 2 (take home) (after the **dropna()** demonstration)

Task: In Section 4.1, a dummy record with a missing value is injected and then removed using the **dropna()** function.

Question: If 40% of documents were missing the **category_name** label, why would simply dropping these rows (with **dropna()**) be a flawed data mining strategy? Name one objective alternative so the text data can still be utilized.

Answer:

1. *It is because the function dropna() deletes every row that has a missing value*
   => So when 40% of documents were missing the category_name label, if using dropna() function will delete all 40% part of the data, while this deleted part can be used for different goal. And also, this also create biased data + less reliable result, as the data becomes smaller would create greater volatility.

2. *Objective alternative: I would use kNN method. Represent every document as a vector (document-term matrix), compute the cosine similarity between each unlabeled document and the labeled documents, take the k most similar labeled documents, and assign their majority label. This is objective because the label is learned from the data itself.*

AI used for this answer: No

Question 3: Data Sampling Risks

Notebook: Section 5.1 — Question 3 (take home)

Task: In Section 5.1, the code uses **X_sample = X.sample(n=1000)** to create a subset. Compare the bar chart of the original category distribution to the sampled distribution.

Questions:

1. Do the exact category proportions perfectly match the original?
2. What is the specific risk of simple random sampling on a highly imbalanced dataset?

Answer:

	soc.religion.christian	**sci.med**	**comp.graphics**	**alt.atheism**
The original (2257)	599 (~26.59%)	594 (26.32%)	584 (~25.88%)	480 (~21.27%)
The sample (1000)	248 (24.8%)	260 (26%)	278 (27.8%)	214 (21.4%)

1. *The differences between the size of before and after taking sample from each category are not too large. This is because the data size at first are roughly similar. However, after taking the sample, the rank between 4 categories has changed.*

2. *There are 2 potential risks of random simple random sampling on a highly imbalanced dataset:*
   +, As I have mentioned on the above, if the original size of each class are much more different. If we take the random sample => the smaller class's sample could be too small, or maybe a model trained on the sample may ignore that class.
   +, The rank of each class could also change.

AI used for this answer: No

Question 4: Text Vectorization and Matrix Sparsity

Notebook: Section 5.3 — Question 4 (take home)

Task: In Section 5.3, the **CountVectorizer** converts text into a document-term matrix using **stop_words='english'**. Change this parameter to **stop_words=None** and re-run the cell to check the shape of **X_counts**.

Questions:

1. Does the total number of features (columns) increase or decrease?
2. Since stopwords are highly common words that appear in almost all documents, what effect does adding them back have on the overall sparsity (the percentage of zeros) of the matrix?

Answer:

	shape	**number of features (columns)**	**sparsity**
stop_words='english'	(2257, 35482)	35482	~99.68%
stop_words=None	(2257, 35788)	35788	~99.55%

1. *Total number of features increases from 35482 -> 35788. The stop words removed by 'english' come back into the vocabulary.*

2. *Adding them back create the decrease in the overall sparsity.*

According to the sparsity formula: ***sparsity = 100.0 * (1 - X_counts.nnz / (X_counts.shape[0] * X_counts.shape[1])).***

+, Since at first, stop_words='english' -> filter nearly 318 common English words, after change it back to stop_words=None, it will bring back those words => number of columns increases. => denominator increases.

+, On the other hand, when change from stop_word='english -> stop_words=None, the number of non zero cells also increases. => numerator increase.

+, However, stop words appear in almost every document, so each added column is mostly non-zero. => the numerator grows much faster than the denominator, so the share of zeros falls => sparsity decreases, but not much.

AI used for this answer: No

Question 5: Feature Selection — Variance Thresholding

Notebook: Section 5.3.1 — Question 5 (take home)

Task: In the variance filtering section, re-run **variance_filter** using threshold values **threshold = 0.0, threshold = 0.01, and threshold = 0.05**. Record the number of features marked as **keep == True**.

Questions:

1. What happens to the total size of the vocabulary as the variance threshold increases from 0.0 to 0.05?
2. In terms of document frequency, what kinds of words have variance close to zero and are eliminated first?
3. Why is eliminating near-zero variance features considered a safe way to reduce dimensionality without harming model accuracy?

Answer:

Threshold	**Number of terms kept (keep == True)**
0.0	35482 kept
0.01	5089 kept
0.05	1491 kept

1. *The total size of vocabulary shrinks quickly as the threshold increases from 0.0 to 0.05.*

2. *Those eliminated first with var == 0.000443 are:*
   +, Name: zvonko, zvi, kuala;
   +, Random string: zz, zzz;
   +, Code: c5kxix, c5l184;
   +, Number: 0000, 000406.

3. *Since the var is pretty low ~0 => it rarely appears in different documents, these rare terms are usually names, typos, codes or numbers. If a model relied on them, it would memorize specific documents (overfitting). When the model meets new document => it won't generalize to new documents.*

AI used for this answer: No

Question 6: Feature Correlation — Linear (Pearson) vs. Monotonic (Spearman)

Notebook: Section 5.3.2 — Question 6 (take home)

Task: Run the correlation filter against the binary indicator for the **comp.graphics** class. Inspect the top 10 features sorted by **pearson_r** and compare them to the top 10 features sorted by **spearman_r**.

Questions:

1. Are the top-correlated words generic stop words or domain-specific terminology for graphics?
2. What is the mathematical difference in what Pearson correlation measures versus what Spearman rank correlation measures?
3. In what specific scenario would a word have a high Spearman rank correlation with a topic, but a significantly lower Pearson correlation?

Answer:

1. *Not entirely stop words or domain-specific terms:*
   +, In Pearson are mixed equally between stop words, domain specific terms, and other kinds.
   +, In Spearman most of them are domain specific terms, the rest are other kinds.

2. *Mathematical difference between Pearson and Spearman correlation measure:*

+, Pearson measures the linear relationship between the raw counts and the 0/1 class indicator (covariance divided by the product of the standard deviations), so a few very large counts can change it a lot.

+, Spearman is Pearson computed on the ranks of the values instead of the raw values, so it measures a monotonic relationship (one variable goes up when the other goes up, not necessarily along a straight line) and is less affected by extreme values. For examples: Word "graphics" in Pearson 0.178 (rank 13th), but in Spearman 0.521 (rank 1st)

3. *Since "graphics" appears a lot in comp.graphics => in Spearman graphics ranks high. Additionally, few posts that repeat the word many times cause data points to deviate significantly from a straight line, which pulls the Pearson correlation downward. However, Spearman's correlation only considers the ranks, so it remains unaffected.*

AI used for this answer: No

Question 7: Frequent Pattern Mining — Algorithm & Parameter Dynamics

Notebook: Section 5.4.2 — Question 7 (take home)

Task: In the Frequent Pattern Mining section, run **Top-K FAE** with **k = 100 and k = 500**. Then, run **MaxFPGrowth** using minimum support thresholds **minSup=9 and minSup= 3**.

	Num of patterns	Run time
MaxFPGrowth, minSup = 9	710	~0.17s
MaxFPGrowth, minSup = 3	9547	~3.08s
Top-K FAE, k = 100	100	~0.05s
Top-K FAE, k = 500	500	~1.5s

Questions:

1. How does decreasing **minSup** (from 9 to 3) in **MaxFPGrowth** affect the execution runtime and the total number of discovered patterns?
2. How does the parameter k in **Top-K FAE** differ fundamentally from **minSup** in **MaxFPGrowth** when deciding which patterns to keep?

3. Why are "maximal" frequent patterns structurally more concise than standard frequent patterns?

Answer (1) — Runtime and pattern count when lowering minSup:

1. *Decrease minSup make the number of patterns increase nearly 13 times (710 to 9547), run time increase nearly 18 times (0.17 to 3.08s)*

Answer (2) — Top-K k vs MaxFPGrowth minSup:

2. *Difference between Top-K FAE and minSup are:*
   +, minSup fixes the threshold: every pattern with support ≥ minSup is kept, so the number of patterns is not known in advance (710 vs 9,547 above).
   +, k fixes the number of patterns: Top-K keeps exactly the k patterns with the highest support, and the support threshold is found automatically (it is the support of the k-th pattern). Increasing k from 100 to 500 lowers this implicit threshold, so the runtime grows (0.05 s → 1.5 s).

Answer (3) — Maximal vs standard frequent patterns:

3. *Difference between Maximal and Standard frequent patterns:*
   +, Definition: A maximal pattern is a frequent itemset that has no frequent proper supersets.
   +, Downward closure property: Every subset of a frequent itemset is also frequent. Therefore, a single maximal pattern of size m represents $2^m - 1$ *non-empty subsets, allowing the entire set of frequent patterns to be derived solely from the maximal list. The trade-off is that the maximal list does not keep the support of each subset.*

AI used for this answer: Yes

If yes, how: Check the downward closure property in the 3rd answer

External references (links): https://github.com/UdayLab/PAMI

Question 8: Dimensionality Reduction — Motivation

Notebook: Section 5.5 — Question 8 (take home) (before PCA/t-SNE code)

Task: After text vectorization, the **X_counts** matrix creates tens of thousands of distinct features (words).

Questions:

1. If we want to visually map how these documents group by category on a scatter plot, why is it mathematically and visually impossible to plot the raw **X_counts** matrix directly?
2. What specific category of data mining algorithms (referenced in the helper files) must be applied first?

Answer:

1. *It is impossible because:*
   +, Visually impossible: A scatter plot only has 2 (or 3) axes.
   +, Mathematically: Plotting the raw matrix means choosing 2 columns (2 words) and dropping the other 35,480, which loses almost all of the document's information.

2. *Dimensionality reduction algorithms must be applied first.*
   The helper file helpers/visualization_helpers.py uses PCA, t-SNE and UMAP to project each document onto 2 (or 3) new coordinates that combine information from all words.

AI used for this answer: No

Question 9: Dimensionality Reduction — Linear (PCA) vs. Non-linear Manifolds (t-SNE/UMAP)

Notebook: Section 5.5 — Question 9 (take home) (after 2D/3D plots)

Task: In the 2D/3D Dimensionality Reduction section, compare the scatter plot projection generated by **PCA** against that generated by **t-SNE** or **UMAP**.

Questions:

1. Which method **(PCA or t-SNE/UMAP**) shows distinct, separated cluster islands for the 4 document categories, and which shows heavy overlap around the center?
2. Why does **PCA** struggle to separate high-dimensional text categories compared to **t-SNE** or **UMAP**?
3. Why must data analysts be careful not to over-interpret the physical distance between separate clusters in a **t-SNE** plot?

Answer:

1. +, t-SNE and UMAP show more distinct and separated cluster islands, and UMAP shows this more clearly.

+, PCA shows heavy overlap.

2. *PCA is more affected by outliers compared to t-SNE & UMAP because:*
   +, PCA optimizes the global variance. So if there is some outlier with big amount => easier to be stretched
   +, PCA (Linear): PCA has no notion of local neighborhoods, it focuses on preserving large distances => deduct the cluster structure of nearby points.
   +, t-SNE & UMAP (Non-linear Manifold & Local Neighborhood): Both algorithms construct a probabilistic or adjacency graph based on the distances between each data point and its knn.

3. *In my run, the t-SNE plot in the 1×3 figure (term-document matrix only, no fixed seed) and the t-SNE plot in the 3×3 grid (pattern-augmented data, random_state=42) look similar overall, but sci.med and alt.atheism switch positions, even though the data changed only slightly.*

   ⇨ *Therefore, the size of a cluster and the distance between two clusters in a t-SNE plot carry little meaning; only which points are grouped together can be trusted.*

AI used for this answer: Yes

If yes, how: Cite the definition of PCA and t-SNE and UMAP

External references (links): https://distill.pub/2016/misread-tsne/

Question 10: One-Hot Binarization

Notebook: Section 5.6 — Question 10 (take home)

Task: In Section 3.1, examine **X['category']** (integers 0, 1, 2, 3). Then look at the output of **LabelBinarizer** in Section 5.6, where each category is converted into a binary vector (e.g., [1, 0, 0, 0]).

Questions:

1. How many binary columns are created to represent the 4 categories?
2. If an algorithm (like linear regression or KNN) is given the raw integer column **(0, 1, 2, 3)** as input, what false mathematical assumption does it make about the categories?

3. How does one-hot binarization eliminate this false assumption?

Answer:

1. *There are 4 columns represented, one per category. For example:*

category	**category_name**	**bin_category**
1	comp.graphics	[0, 1, 0, 0]
2	sci.med	[0, 0, 1, 0]
3	soc.religion.christian	[0, 0, 0, 1]

2. *If the raw integer column is given as an input, the distance of each categories will be ranked according to the name (follow the alphabetical order of the names) not based on its content.*

3. *One-hot binarization eliminates this false assumption by giving each category its own column, so the columns are independent and no category is larger than another. Every pair of categories differs in exactly two positions, so all pairs have the same Euclidean distance. A linear model also gets a separate weight for each category instead of one shared coefficient.*

AI used for this answer: No
